

MATH 2270 — Linear Algebra

Study Guide for Test 3 — Unit 3

Unit 3: Eigenvalues, Diagonalization, Markov Chains (§5.1–5.3, 5.5) and Inner Products, Orthogonality, Least Squares (§6.1–6.5, 7.1)

What to expect on Test 3: 50 minutes, 100 points, six sections — True/False, Multiple Choice, Definitions, Quick Computations (no software), an “Inner Product or Not?” question, and a Computation section where a row-reduction calculator or software *is* allowed. The test covers Days 30–40 (Network Day is not tested; spectral decomposition is not tested).

How to use this guide: for each topic read the vocabulary and key facts, study the worked example, then try the practice prompts before checking the bracketed answers. Then take Practice Test 3 under real time pressure (it is twice as long as the real test, so do half of each section in 50 minutes), and drill weak spots with the Unit 3 Super Review deck.

§5.1 / 5.3 — Eigenvalues and Eigenvectors

Key vocabulary

- Eigenvector: a *nonzero* $\mathbf{v}$ with $A\mathbf{v} = \lambda\mathbf{v}$; the scalar λ is its eigenvalue.
- Characteristic polynomial: $\det(A - \lambda I)$. Its roots are the eigenvalues.
- Eigenspace of λ : all solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ (a subspace).

What you need to know

- Find eigenvalues first (solve $\det(A - \lambda I) = 0$), then for each λ row reduce $A - \lambda I$ to find the eigenvectors.
- For a triangular matrix the eigenvalues are the diagonal entries.
- A is invertible $\iff 0$ is not an eigenvalue. The determinant is the product of the eigenvalues; the trace is their sum.
- Real matrices can have complex eigenvalues, and they come in conjugate pairs (a rotation has no real eigenvectors).
- Any nonzero multiple of an eigenvector is again an eigenvector.

Worked example. Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$. $\det(A - \lambda I) = (1 - \lambda)(-2 - \lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3)$. For $\lambda = 2$: $A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$, eigenvector $(2, 1)$. For $\lambda = -3$: $A + 3I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$, eigenvector $(1, -2)$.

Try it yourself.

- Eigenvalues of $\begin{bmatrix} 3 & 0 & 0 \\ 1 & -2 & 0 \\ 5 & 4 & 7 \end{bmatrix}$. [**3**, -2, **7**]
- A 3×3 matrix has eigenvalues 2, -1, 4. What is $\det A$? Is it invertible? [-8; **yes**]
- Is $(1, 2)$ an eigenvector of $\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$? [**No**: $A\mathbf{v} = (2, 7)$]

§5.2 — Diagonalization**Key vocabulary**

- Similar: $B = P^{-1}AP$ for some invertible P . Similar matrices share eigenvalues (and determinant, trace) but *not* eigenvectors.
- Algebraic multiplicity: how many times λ is a root. Geometric multiplicity: $\dim$ of its eigenspace.
- Defective: a matrix with too few independent eigenvectors (not diagonalizable).

What you need to know

- A is diagonalizable $\iff$ it has n linearly independent eigenvectors $\iff$ geometric = algebraic multiplicity for every eigenvalue.
- n distinct eigenvalues $\Rightarrow$ diagonalizable (but repeated eigenvalues do not rule it out).
- $A = PDP^{-1}$: columns of P are eigenvectors, D holds the matching eigenvalues, in the same order.
- $A^k = PD^kP^{-1}$, and D^k just raises each diagonal entry to the k .
- Geometric multiplicity is always at most algebraic multiplicity.

Worked example. Diagonalize $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$. From the previous example, $P = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$ and $D = \text{diag}(2, -3)$, so $A = PDP^{-1}$. Then $A^3 = P \text{diag}(8, -27) P^{-1}$. Quick check: $AP = PD$.

Try it yourself.

- Is $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ diagonalizable? [**No**: $\lambda = 3$ has algebraic multiplicity 2, geometric 1]
- If $D = \text{diag}(2, -1)$, find D^5 . [**diag**(32, -1)]
- A is 3×3 with eigenvalues 1, 1, 4 and a 2-dimensional eigenspace for $\lambda = 1$. Diagonalizable? [**Yes**]

§5.5 — Markov Chains**Key vocabulary**

- Stochastic matrix: nonnegative entries and every *column* sums to 1. Probability vector: nonnegative entries summing to 1.
- State vectors: $\mathbf{x}_{k+1} = P\mathbf{x}_k$, so $\mathbf{x}_k = P^k\mathbf{x}_0$.
- Regular: some power of P has all positive entries. Steady-state vector: $P\mathbf{q} = \mathbf{q}$ with entries summing to 1.

What you need to know

- Every stochastic matrix has $\lambda = 1$ as an eigenvalue, so a steady state is an eigenvector for $\lambda = 1$.
- To find $\mathbf{q}$: row reduce $(P - I)\mathbf{q} = \mathbf{0}$, then scale so the entries sum to 1 (not length 1).
- For a regular chain, $P^k \mathbf{x}_0 \rightarrow \mathbf{q}$ no matter what $\mathbf{x}_0$ is.

Worked example. $P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$, $\mathbf{x}_0 = (1, 0)$. Then $\mathbf{x}_1 = P\mathbf{x}_0 = (0.8, 0.2)$. Steady state: $(P - I)\mathbf{q} = \mathbf{0}$ gives $-0.2q_1 + 0.3q_2 = 0$, so $q_1 : q_2 = 3 : 2$ and $\mathbf{q} = (3/5, 2/5)$.

Try it yourself.

- Is $\begin{bmatrix} 0.5 & 0.4 \\ 0.5 & 0.7 \end{bmatrix}$ stochastic? [**No: column 2 sums to 1.1**]
- Find the steady state of $\begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$. $[(2/3, 1/3)]$

§6.1 — General Inner Products**Key vocabulary**

- Inner product axioms: symmetry, linearity, and positivity ($\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $= 0$ only for $\mathbf{v} = \mathbf{0}$).
- Norm $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$; distance $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
- Examples: weighted dot product $\sum w_i u_i v_i$ ($w_i > 0$), $\text{tr}(U^T V)$ on matrices, $\int_a^b f g dx$ on $C[a, b]$.

What you need to know

- To show something is *not* an inner product, find ONE failed axiom with a specific counterexample (usually positivity: a nonzero vector with $\langle \mathbf{v}, \mathbf{v} \rangle \leq 0$).
- Negative weights, or a weight of 0, break positivity. A cross term can also break it; complete the square to check.
- Norms and distances change when the inner product changes.

Worked example. On $\mathbb{R}^2$ let $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1 v_1 + 5u_2 v_2$, $\mathbf{u} = (1, 2)$, $\mathbf{v} = (3, -1)$. Then $\langle \mathbf{u}, \mathbf{v} \rangle = 6 - 10 = -4$ and $\|\mathbf{u}\| = \sqrt{2 + 20} = \sqrt{22}$. On $C[0, 1]$: $\langle x, x^2 \rangle = \int_0^1 x^3 dx = \frac{1}{4}$ and $\|x\| = \sqrt{1/3}$.

Try it yourself.

- Is $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_2$ an inner product? [**No: $\langle (0, 1), (0, 1) \rangle = -1 < 0$**]
- On $C[0, 1]$, find $d(x, 1)$. $[\sqrt{\int_0^1 (x - 1)^2 dx} = 1/\sqrt{3}]$

§6.2 — Orthogonality in Inner Product Spaces

Key vocabulary

- $\mathbf{u} \perp \mathbf{v}$ means $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for the inner product being used.
- Orthogonal complement $W^\perp$: all vectors orthogonal to every vector in W .
- Orthogonal set: pairwise orthogonal. Orthonormal: orthogonal and all unit length.

What you need to know

- Cauchy–Schwarz: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
- Pythagoras: $\mathbf{u} \perp \mathbf{v} \Rightarrow \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
- Orthogonal sets of nonzero vectors are linearly independent. $W \cap W^\perp = \{\mathbf{0}\}$ and $\dim W + \dim W^\perp = n$.
- To find $W^\perp$ in $\mathbb{R}^n$: put a spanning set of W as the rows of a matrix and find its null space.
- With an orthonormal basis, the coordinates are $c_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$.

Worked example. With $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 5u_2v_2$, are $(1, 1)$ and $(5, -2)$ orthogonal? $\langle \mathbf{u}, \mathbf{v} \rangle = 10 - 10 = 0$, so yes — even though their *dot* product is $3 \neq 0$. In $\mathbb{R}^3$, let $W = \text{span}\{(1, 1, 0), (0, 1, 1)\}$; $W^\perp$ is the null space of $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, spanned by $(1, -1, 1)$.

Try it yourself.

- Under $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2$, find k so $(3, 1) \perp (1, k)$. [$k = -1$]
- If $\|\mathbf{u}\| = 1$, $\|\mathbf{v}\| = 4$ and $\mathbf{u} \perp \mathbf{v}$, find $\|\mathbf{u} + \mathbf{v}\|$. [$\sqrt{17}$]

§6.3 — Gram–Schmidt and QR

Key vocabulary

- Gram–Schmidt: turns independent $\mathbf{v}_1, \dots, \mathbf{v}_k$ into orthogonal $\mathbf{u}_1, \dots, \mathbf{u}_k$ with the same span.
- QR decomposition: $A = QR$, Q with orthonormal columns, R upper triangular.

What you need to know

- $\mathbf{u}_1 = \mathbf{v}_1$; $\mathbf{u}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1$; $\mathbf{u}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 - \frac{\langle \mathbf{v}_3, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2$.
- Use whatever inner product the problem gives, in *both* the numerators and the denominators.
- Normalize at the end: $\mathbf{q}_i = \mathbf{u}_i / \|\mathbf{u}_i\|$. Then $Q = [\mathbf{q}_1 \cdots \mathbf{q}_k]$ and $R = Q^T A$.
- Checks: $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$, and $Q^T Q = I$.

Worked example. Apply Gram–Schmidt (dot product) to $\mathbf{v}_1 = (1, 1, 0)$, $\mathbf{v}_2 = (1, 0, 1)$. $\mathbf{u}_1 = (1, 1, 0)$; $\mathbf{v}_2 \cdot \mathbf{u}_1 = 1$, $\mathbf{u}_1 \cdot \mathbf{u}_1 = 2$, so $\mathbf{u}_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (\frac{1}{2}, -\frac{1}{2}, 1)$. Check: $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$.

Try it yourself.

- Normalize $\mathbf{u}_1$ above. [$\frac{1}{\sqrt{2}}(1, 1, 0)$]
- In $A = QR$, what does the $(1, 1)$ entry of R equal? [$\|\mathbf{a}_1\|$]

§6.4 / 6.5 — Least Squares

Key vocabulary

- Least-squares solution $\hat{\mathbf{x}}$: minimizes $\|\mathbf{b} - A\mathbf{x}\|$. Residual: $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$.
- Normal equations: $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. Design matrix: ones column plus the data columns.

What you need to know

- Normal equations are always consistent; the solution is unique $\iff$ the columns of A are independent.
- The residual is orthogonal to $\text{col}(A)$: $A^T \mathbf{r} = \mathbf{0}$.
- Line fit $y = a + bx$: rows $(1, x_i)$. Parabola: rows $(1, x_i, x_i^2)$. $\text{SSE} = \|\mathbf{r}\|^2$.
- Set up $A^T A$ and $A^T \mathbf{b}$ by hand, then use the row reducer to solve.

Worked example. Fit a line to $(0, 2), (1, 3), (2, 7)$. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} = (2, 3, 7)$. $A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^T \mathbf{b} = (12, 17)$. Solving gives $y = 1.5 + 2.5x$. Residuals $(0.5, -1, 0.5)$ sum to 0, as they must.

Try it yourself.

- Predict y at $x = 3$ for that line. [9]
- What is $A^T A$ for the points $(1, \cdot), (2, \cdot), (3, \cdot)$ with a line fit? $\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$

§7.1 — Orthogonal Matrices

Key vocabulary

- Orthogonal matrix: a *square* matrix Q with $Q^T Q = I$, equivalently $Q^{-1} = Q^T$.

What you need to know

- The columns (and the rows) of an orthogonal matrix form an orthonormal basis.
- $\|Q\mathbf{x}\| = \|\mathbf{x}\|$, and Q preserves dot products and angles.
- $\det Q = \pm 1$, and the product of orthogonal matrices is orthogonal.
- A non-square matrix with orthonormal columns has $Q^T Q = I$ but $Q Q^T \neq I$.

Worked example. Is $Q = \frac{1}{13} \begin{bmatrix} 5 & 12 \\ -12 & 5 \end{bmatrix}$ orthogonal? Each column has length $\sqrt{25 + 144}/13 = 1$ and the columns have dot product $\frac{60 - 60}{169} = 0$, so $Q^T Q = I$ and $Q^{-1} = Q^T$. Also $\det Q = 1$.

Try it yourself.

- Is $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ orthogonal? How could you fix it? [No; divide by $\sqrt{2}$]
- If Q is orthogonal and $\|\mathbf{x}\| = 7$, what is $\|Q\mathbf{x}\|$? [7]